

Tutorial 2: Power Electronic Devices and Figures of Merit
EE 3203: Power Electronics

1. A 230 V, single-phase ac supply powers a diode bridge rectifier as shown. The conduction time of each diode is 1 ms in every fundamental cycle. The output dc voltage is filtered with a capacitor C. The load current is a steady dc of 1 A. At steady state,
 - (a) Sketch the output voltage $v_o(t)$ and evaluate the output peak-to-peak voltage ripple.
 - (b) Sketch the capacitor current $i_C(t)$.
 - (c) Sketch the rectifier output current $i_R(t)$.
 - (d) Evaluate the value of capacitor C.

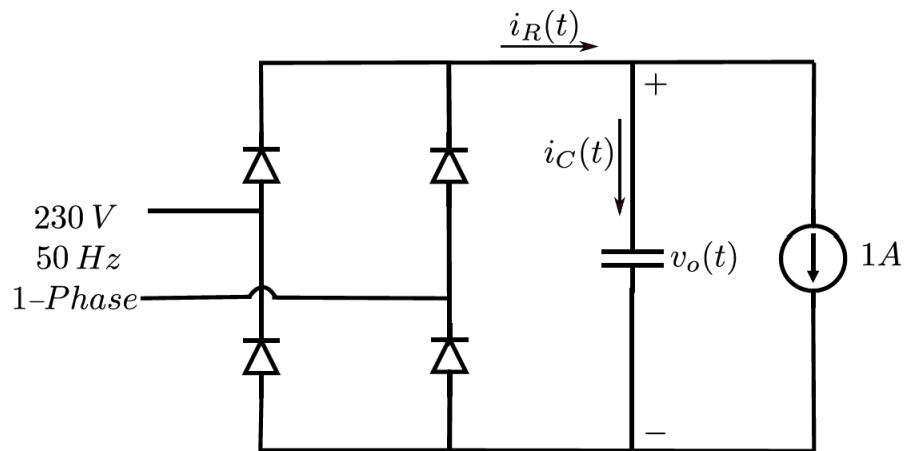


Fig. 1: Single-phase bridge rectifier.

2. In the boost converter shown, the output voltage $V_o = 30V$ for a duty ratio of 0.5.
[Duty ratio of 0.5 means the switch spends equal length of time in positions 1 and 2.]

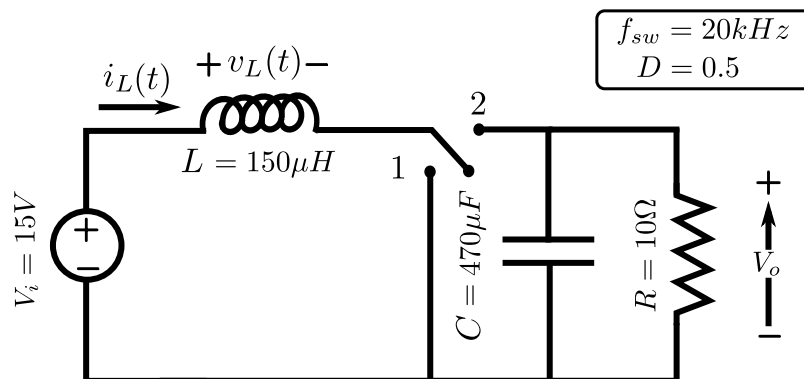


Fig. 2: Boost converter.

- (a) Draw the equivalent circuit for switch positions 1 and 2.

- (b) Using the equivalent circuits, at steady state, sketch the voltage $v_L(t)$ and current $i_L(t)$. Also, compute the peak-to-peak inductor current ripple. [Hint: Assuming no losses, the average power at the input and output can be equated to compute the average value of $i_L(t)$.]
 - (c) Using the equivalent circuits, at steady state sketch the current through the capacitor C. Also, compute the peak-to-peak voltage ripple.
3. The input voltage $v(\omega t)$ and input current $i(\omega t)$ of a single phase half-controlled bridge rectifier feeding a highly inductive load (constant output current) is shown in the figure. The input current is periodic with the same period as the input voltage.

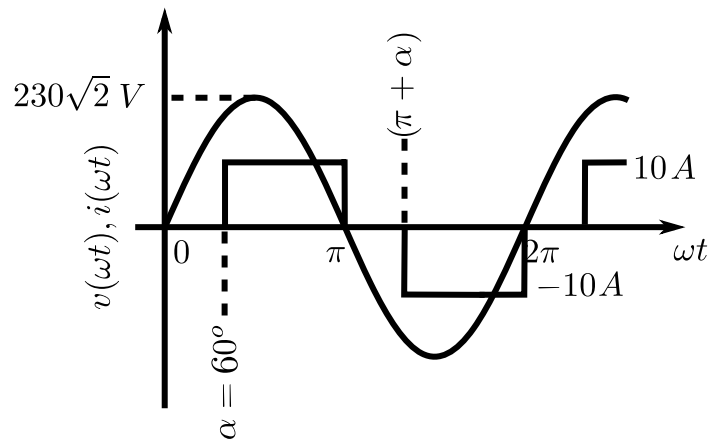


Fig. 3: Input voltage and current in a single-phase half-controlled bridge rectifier.

- (a) Compute the rms value of the input current.
 - (b) Compute the rms value of the fundamental frequency component of input current.
 - (c) Compute the crest factor of the input current.
 - (d) Compute the displacement factor and power factor.
 - (e) Compute the input average power as $P_{avg} = \frac{1}{2\pi} \int_0^{2\pi} v(\omega t)i(\omega t)d(\omega t)$.
 - (f) Compute the power due to the fundamental frequency component of voltage ($V_{1,rms}$) and current ($I_{1,rms}$) as $P_1 = V_{1,rms}I_{1,rms} \cos \phi_1$. Compare P_1 with P_{avg} . Observe that since the voltage is purely sinusoidal, the total average power results only from the interaction of the fundamental frequency components of voltage and current. The other components do not contribute to the power transfer and lead to unwanted power loss in the transmission/distribution lines.
4. The electrical model of an SCR and two periodic operational waveforms of the device are shown. The first waveform is a half sinusoid with a peak current of 100 A. The second waveform is a pulse width modulated waveform with a peak of 100 A and a duty ratio of 50%. The power dissipation in the device is respectively 56.8 W and 100 W under the two operating conditions.

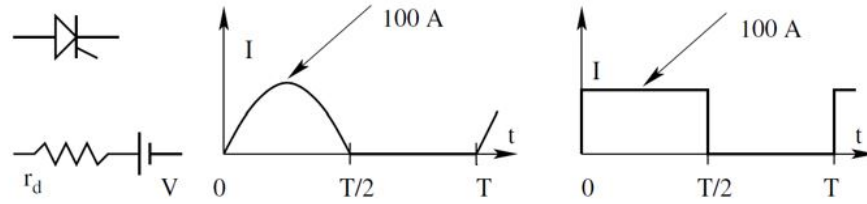


Fig. 4: Electrical model of SCR and current waveforms.

- (a) Evaluate the average and rms current for the two waveforms.
 - (b) Evaluate model parameters V and r_d of the SCR.
5. The switching voltage and current waveforms of a power switching device with a reactive load is shown in the figure.

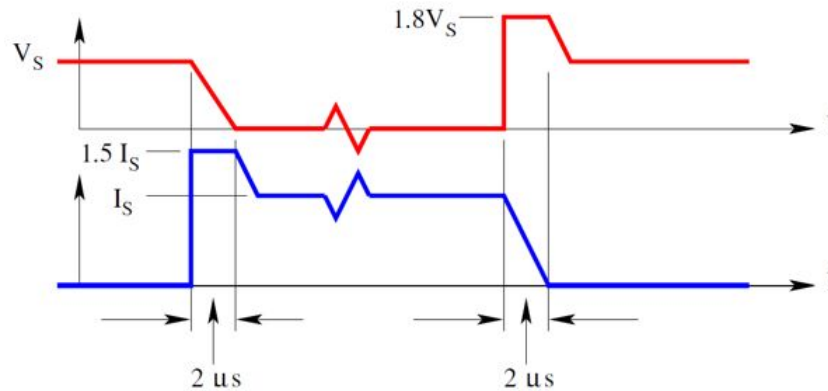


Fig. 5: Switching waveforms of a power switching device.

The voltage V_S and current I_S are respectively 200 V and 50 A. The device exhibits overcurrent during turn-on and overvoltage during turn-off as shown in the figure. The switching frequency is 10 kHz.

- (a) Sketch the device power dissipation as a function of time.
- (b) Evaluate the peak power dissipation during the turn-on transition and the turn-off transition.
- (c) Evaluate the turn-on energy loss, turn-off energy loss and the average switching power loss in the switching device.

Observe that the peak switching power dissipation is quite high compared to the average switching power loss. The peak switching loss and the losses during the switching transitions can cause momentary elevation in the junction temperature of the device and needs consideration in the thermal design. The thermal time constants at the wafer level is low.

- (d) Assuming the switching transition times to be $1\mu s$ instead of $2\mu s$, recompute the turn-on energy loss, turn-off energy loss and the average switching power loss in the switching device. Observe the effect of lower switching transition time duration on the switching loss of the device. Understand the advantage of having a lower switching transition time.

6. Assume $v(t) = \sum_{n=1}^{n=N} V_n \sin(n\omega t - \phi_n)$. Prove, $V_{rms}^2 = \sum_{n=1}^{n=N} V_{n,rms}^2$.

Hint:

$$V_{rms}^2 = \frac{1}{2\pi} \int_0^{2\pi} v^2(t) d(\omega t).$$

$$V_{rms}^2 = \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=1}^{n=N} V_n \sin(n\omega t - \phi_n) \sum_{n=1}^{n=N} V_n \sin(n\omega t - \phi_n) d(\omega t).$$

The multiplication results in a set of product terms with identical n and a set of product terms with non-identical n . The ones with non-identical n can be observed to result in zero on integrating. The other terms result in the square of the individual rms components.